ER PELVIS

STATION: INGUINAL CANAL

The spermatic cord represents the blood vessels, innervation, vas deferens and lymphatics of the testis that descend from the intra-abdominal region to the scrotum. It becomes covered by layers derived from the abdominal wall through which it passes. The layers and the contents are listed below:

1. External spermatic fascia ( ), a soft fibrous covering that the cord derives from the edges of the superficial ring. Hence this layer will only be found on the cord from the external ring onwards.
2. Cremasteric muscle and fascia ( ), derived from the lowermost fibers of the internal oblique muscle and fascia.
3. Internal spermatic fascia ( ), derived from the fascia transversalis of the transversus abdominis muscle. This forms the innermost wrapping around the contents of the cord.
4. The vas deferens or ductus deferens ( ). This duct carries spermatozoa from the testis to the urethra for ejaculation. It has a very characteristic 'tactile' feel (chalky)( ) and, on close inspection, students may be able to identify its artery traveling alongside it.
5. The testicular artery and pampiniform plexus of veins ( ). The plexus of veins courses around the artery and extends up to the deep ring. The lymphatic vessels are too small to be seen by the naked eye.
6. The ilioinguinal nerve lies external to the cremasteric muscle and underneath the external spermatic fascia. It is considered to be part of the content of the canal.
7. Examine the cut surface of the testis ( ). Identify the outer tough capsule known as the tunica albuginea - that surrounds the testis, except posteriorly.
8. Examine the scrotum and verify that its wall does not contain a subcutaneous fatty layer as elsewhere. Instead it is replaced by a smooth muscle layer known as the dartos layer.

Station: Urinary bladder, prostate, and seminal vesicles

The urinary bladder is a muscular sac that stores urine. Its filling capacity is up to about 500 ml or until voiding. Its shape changes from pyramidal when empty to an ovoid mass when full. Located just behind the symphysis pubis, the bladder sits atop the prostate in males and on the pelvic floor in females. In the specimen, examine the following features:

1. The urinary bladder has an upper surface that is covered by peritoneum ( ). Anteriorly, the apex of the bladder sits just behind the pubic symphysis ( ). Opposite this apex is the base ( ). There are two inferolateral surfaces related to the levator ani muscle ( ). Neither surface is covered by peritoneum.
2. The inside of the bladder has numerous folds when empty but becomes smooth upon filling ( ).
   The part of the bladder opposite the base is always smooth ( ) . This area is known as the trigone due to its triangular shape. It has an opening in each angle: two ureteric openings situated at its superolateral angles, and the internal urinary meatus at its tip ( ).
3. The ureter travels along the side wall of the pelvis after entering the pelvis anterior to the sacroiliac joint, and crossing the bifurcation of the common iliac artery. As it heads for the ischial spine, it turns sharply forward towards the bladder. The ureter enters the bladder at its corner ( ), where the vas deferens crosses superiorly.
4. The prostate gland, one of the male internal genital organs, sits inferior to the bladder in the male ( ). It may appear like an inverted pyramid with the apex downwards and the base upwards ( ). The first part of the urethra starts at the neck of the bladder and passes through the prostate gland. The two organs are intimately connected and share the same smooth muscle coat. The posterior aspect of the gland shows a midline groove that is situated between two gently rounded swellings, the posterior lobes of the gland ( ).

5. In males, the two seminal vesicles are related to the base of the bladder ( ). A seminal vesicle is an elongated, lobulated sac lying obliquely on the bladder base, which contributes a watery secretion to the seminal fluid. In females, the base of the bladder is related to the vagina.
6. Also related to the bladder base are the left and right vas deferens ( ) that convey sperm from the testis. Each has a minor dilation at the end which forms the ampulla of the vas deferens ( ). From the ampulla, the vas deferens then joins with the duct of the ipsilateral seminal vesicle to form the ejaculatory duct ( ). These pass through the prostate to open into the urethra posteriorly. These two ducts, together with the urethra, divide the prostate into several lobes.

Station: Uterus and adnexae

The uterus is the principal female internal genital organ. It is situated in the middle of the pelvic cavity between the urinary bladder anteriorly and the rectum posteriorly. It is covered by peritoneum on the anterior and posterior aspects (with more covering posteriorly). From there, the peritoneum continues on to cover the rectum (posteriorly) and urinary bladder (anteriorly) forming the recto-uterine and the utero-vesical pouch, respectively. In the specimen, try to identify the following structures:

1. The fundus of the uterus ( ), which represents the uppermost, rounded part of the organ. It is situated above the entrance to the uterine (Fallopian) tubes.
2. The body of the uterus ( ) is found below the entrance of the fallopian tubes and above the isthmus.
3. The isthmus is where the body becomes narrowed to form a waist ( ).
4. Below this is the muscular sphincter, the cervix ( ), which dilates during pregnancy to allow passage of the fetus.
5. The cervix protrudes through the anterior vaginal wall into the vaginal lumen ( ). It forms recesses between itself and the vaginal walls, known as fornices (singular fornix). The deepest, and most clinically important, is the posterior vaginal fornix ( ). The internal os opens into the uterine cavity ( ) (superiorly). The external os opens into the vagina (inferiorly).
6. The uterine (Fallopian) tubes or oviducts are a pair of muscular tubes attached to the sides of the upper part of the uterus ( ). They are covered by peritoneum in the form of a sleeve. This peritoneum forms the broad ligament of the uterus ( ).
7. Each tube has a fimbriated end ( ) that is near its ovary.
8. This is continuous with the dilated part of the tube, the ampulla ( ). This is the most common site of fertilization.
9. Moving medially, a narrow isthmus ( ) follows and becomes the intramural part, which travels a short distance through uterine wall.
10. The ovaries are suspended from the posterior leaf of the broad ligament by the mesovarium ( ). They also have a suspensory ligament from the side wall of the pelvis, through which travels vessels, nerves ( ), and round ligament. This ligament goes to the corner of the uterus, then travels through the inguinal canal to the labium majus as the round ligament of the uterus ( ).
11. The uterus, if seen from the side, will appear to form a 90 degree angle with the vagina ( ). This angle is referred to as anteverted. It is also slightly bent on the cervix to form an angle of anteflexion ( ). These two angles bring the uterus to lie on the upper surface of the bladder when both are empty.
12. The vagina is a muscular tube that extends from the cervix above to the vulva below ( ) after passing through the gap between the two levator ani muscles. Note the anterior and a posterior wall. The ureter crosses under the uterine artery just lateral to the vaginal cervix.

ER PERINEUM

The perineum is the anatomical region of the pelvis that lies below the pelvic diaphragm. It is a relatively small region yet a clinically important region. Its bony and ligamentous borders give it a diamond outline. It is divided by an imaginary transverse line into an anterior urogenital triangle and a posterior anal triangle. The anal triangle is identical in structure in both sexes. However there are differences regarding the urogenital triangle due to different external genitalia.

Station: Ischioanal Fossa

This is a fat-filled space on either side of the anal canal. In the specimen locate the following:

1. The lateral wall of the fossa, which is formed by the ischium, obturator internus muscle and its fascia ( ).
2. The medial wall, formed by the lateral wall of the anal canal and its sphincter ( ).
3. The posterior wall, consisting of the sacrotuberous ligament overlapped by the gluteus maximus ( ).
4. The anterior wall, consisting of the perineal body and the transverse perineal muscles ( ). The fossa extends anteriorly above these muscles as far as the pubic bone but does not communicate with the opposite side ( ).
5. The roof, consisting of the sloping undersurface of the levator ani muscle ( ).
6. The fossa is filled with fat ( ), and high in the roof are the inferior rectal branches of the internal pudendal nerve and vessels ( ).

Station: Superficial perineal pouch

This is the potential space deep to the membranous layer of superficial fascia. Its posterior limit is the perineal body. Its contents are:

1. The two crura of the penis (or clitoris) attaching to the ischiopubic rami ( ). These are covered by the ischiocavernosus muscles ( ).
2. The bulb of the penis (or clitoris) ( ) covered by the bulbospongiosus muscle ( ) that is attached posteriorly to the perineal body.
3. The superficial transverse perineal muscles that arise from the medial aspects of the ischial tuberosities. These insert in the midline to help form the perineal body or the central tendon of the perineum ( ).
4. The upper limit of the pouch is the undersurface of the urogenital diaphragm ( ). In the female the pouch also includes the labia, vulva and the Bartholin glands and in the male includes the scrotum and the penis.

Also identify the:

5. Body of the penis ( ) and its glans ( ), which relates to the opening of the external urinary meatus.
6. Foreskin, or prepuce, and the frenulum of the prepuce ( ). The penis internally is formed by two corpora cavernosa ( ) and one corpus spongiosum ( ) that contains the urethra ( ). All are surrounded by a thick fascia known as the deep penile fascia ( ). The penis has a vein, two arteries and two nerves. They are located in that order from medial to lateral, deep to this fascia.
7. The female external genitalia comprise two hair-bearing labia majora ( ), and two hairless labia minora located medial to these ( ). The labia majora form the outer boundary of the vulva ( ) while the labia minora form the outer boundary of the vestibule of the vulva ( ). In relation to the vestibule, the vagina opens posteriorly ( ) and the urethra anteriorly ( ). Anterior to the urethral opening is the clitoris ( ).

Station: Deep Perineal Pouch

This is a potential space lying above the urogenital diaphragm and below the fascia that covers it from above. The urogenital diaphragm is a membrane that attaches to the medial aspects of the ischiopubic rami and closes the gap between them.

1. The pouch contains the deep transverse perineal muscle that resemble the superficial counterparts ( ).
2. The sphincter urethrae muscle anterior to the above muscle ( ), the external urethral sphincter.

The membranous part of the urethra ( ). In the male are also included the Cowper's glands (bulbo-urethral glands)